

Madhav Umesh Kamble

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EDUCATION

Indian Institute of Information Technology Allahabad

M.Tech in Information Technology (Data Engineering)

- CGPA: 8.46 / 10
- Research: *Enhancing Genetic Algorithm with Explainable AI for Last-Mile Routing*

SKN Sinhgad College of Engineering

B.Tech in Computer Science and Engineering

- CGPA: 9.31 / 10

Prayagraj, Uttar Pradesh

Aug. 2025 – July 2027

Korti, Pandharpur

Aug. 2021 – May 2025

EXPERIENCE

Leadsoft IT Solutions

Full Stack Web Development Intern

Virtual

Aug. 2023 – Oct. 2023

- Built a Gate Pass Management mobile app (React Native + MERN stack) supporting digital pass requests, multi-level approvals, and real-time tracking.
- Developed RESTful APIs in Node.js/Express with MongoDB for pass lifecycle management and real-time status updates.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript, TypeScript, Java, C, C++, HTML, CSS

LLM & GenAI: OpenAI API, Groq API (Llama 3.3 70B), Ollama, RAG, Multi-Agent Orchestration, Vector Embeddings, Pydantic Structured Outputs

Data & ML: NumPy, Pandas, Scikit-learn, XGBoost, SHAP, SciPy, PyTorch, Numba, Matplotlib, Seaborn

Backend & Databases: FastAPI, Node.js, Express.js, PostgreSQL, pgvector, SQLite, ChromaDB, MongoDB

Web: React.js, Vite, Tailwind CSS

Tools & Platforms: Git, GitHub, Docker, Linux, VS Code, GitHub Actions, pytest

Relevant Coursework: Big Data Analytics, Data Visualisation, Machine Learning, Deep Learning, OS, CN, DBMS

PROJECTS

Autonomous Data-Analyst Agent — [GitHub](#) — [Live Demo](#)

Feb. 2026 – July 2026

FastAPI, React, PostgreSQL, Groq (Llama 3.3 70B), RAG, pgvector

- Built a multi-agent NL-to-SQL system (Planner → RAG → SQL Generator → Executor → Critic → Summarizer) with a bounded self-correction loop capping cost at 8 LLM calls/query.
- Enforced SQL safety via a database-level read-only role as the real security boundary, verified by tests that attack the grant directly; backed by a syntax pre-check and query guards.
- Built a dual-mode RAG layer (pgvector dev / Postgres full-text prod) swappable via config; validated with 67 automated tests in CI, deployed on a zero-cost stack.
- Deployed across three free-tier platforms (Neon, Render, Vercel) with a monitored keep-warm health check, behind a strict-TypeScript React 19 frontend with a cold-start detection hook and a reasoning-trace panel mirroring the backend contract field-for-field.

ClaimSense – AI Claims Triage & Policy Q&A Assistant — [GitHub](#) — [Live Demo](#)

Apr. 2026 – June 2026

Python, OpenAI API (GPT-4o-mini), Pydantic, ChromaDB (RAG), SQLite, Docker

- Built an AI-assisted claims triage system combining LLM classification/summarization with a RAG pipeline (ChromaDB) grounding policy Q&A in real documents, behind a deterministic rules-plus-AI-confidence routing engine for fully auditable decisioning.
- Implemented a governed PII-redaction layer with a single enforced audit-log choke point, plus a confidence-calibration module empirically verifying LLM self-reported confidence scores are statistically trustworthy (weighted calibration error 0.038).
- Ran an adversarial self-audit beyond the existing 21-test suite, uncovering and fixing 6 real bugs – including a silent data-loss bug in the persistence layer and a validation-bypass entry point.

Certified-Optimal Memetic Search for Driver-Imitating Last-Mile Routing

Jan. 2026 – Present

Python, Genetic Algorithms, XGBoost, SHAP, Held-Karp DP, Numba, PyTorch, SciPy

- Audited and repaired a published XAI-enhanced genetic algorithm (Kim, Khir & Lee, IEEE TEVC 2026), finding a missing normalization that left its core explainability mechanism inert ($\sim 133\times$ objective-scale imbalance), recovering a corrected baseline of 0.0439 on Amazon's 3,052-route Last-Mile Routing Challenge.
- Proved via paired-Wilcoxon ablation ($p=0.82$) that the paper's trained SHAP pipeline could be replaced by a closed-form formula, then built a memetic GA with an exact Held-Karp DP local search improving the baseline 11.6% ($0.0439 \rightarrow 0.0388$), surpassing Amazon's public LMRRC leaderboard 3rd-place result (0.0391).

ACHIEVEMENTS & CERTIFICATIONS

Competitive Programming: LeetCode Rating 1669 | CodeChef Rating 1441

Runner-Up – Sinhgad Hackathon 2023 and 2022

Certifications: 0 to 100 Full-Stack Cohort (100xDevs, 2024) | Cyber Threat Intelligence 101 (arcX, 2024)